

Rachel Malea Russ

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EDUCATION

Purdue University, West Lafayette, Indiana

Bachelor's of Science in Aeronautical and Astronautical Engineering

Minors: Mathematics and Economics

May 2027

GPA 3.66/4.00

Relevant Coursework:

- *Fluid Dynamics, Aerodynamics, Control System Analysis, Signal Analysis, Partial Differential Equations, Dynamics and Vibrations, Statics, Structural Analysis, Flight Dynamics and Controls, Introduction to Propulsion, Computational Combustion and Propulsion, and Thermodynamics*

Brentwood High School, Brentwood, Tennessee

Honors: Summa Cum Laude, GPA 4.63 weighted, 4.0/4.0 unweighted

May 2023

EXPERIENCE

Research Intern, **University of Tennessee Space Institute**, Tullahoma TN

June 2026 - Current

- Investigate hypersonic inlet unstart using computational fluid dynamics (CFD) under the supervision of Dr. Esteban Cisneros-Garibay
- Execute high-fidelity simulations using the PeleC compressible flow solver on high-performance computing (HPC) clusters
- Adapt PeleC source code and simulation parameters to investigate effects of operating conditions, numerical schemes, and mesh refinement strategies
- Post-process large-scale simulation datasets using Python and VisIt to analyze shock interactions, flowfield evolution, and inlet performance
- Assess adaptive mesh refinement (AMR) behavior and numerical stability to improve solution accuracy and diagnose numerical instabilities
- Communicated research findings through technical presentations and scientific visualizations

Undergraduate Researcher, **Purdue University**, West Lafayette IN

February 2026 - May 2026

- Develop compressible CFD models in Ansys Fluent to investigate shock-wave propagation and steady shock behavior under the supervision of Dr. Haifeng Wang
- Analyze pressure, velocity, and shock structure data to characterize compressible flow behavior

Propulsion/ Fluids Engineer, **Purdue Space Program (PSP)**, West Lafayette IN

August 2024 - May 2026

- Supported fluid system design for an entry in the International Lander Challenge
- Designed waterline routing and produced piping and instrumentation diagrams (P&IDs) for fluid system documentation
- Contributed to rocket tank, diffuser, and ground support equipment (GSE) design

Intern, **Vanderbilt LifeFlight**, Nashville TN

June 2025 - August 2025

- Developed an HTML/JavaScript application that displayed real-time helipad turbulence conditions for flight operations, improving operational awareness for pilots and dispatch personnel
- Integrated wind tunnel data from RWDI with established operational turbulence thresholds
- Collaborated with engineering and aviation personnel to improve flight safety decision-making

Experimental Fluid Dynamics and Aerodynamics Laboratory,
Purdue University, West Lafayette IN

Spring 2025, Fall 2025

- Operated subsonic and supersonic wind tunnels to conduct aerodynamic and fluid dynamics experiments
- Processed experimental data to evaluate flow behavior and aerodynamic performance
- Designed, proposed, and executed two team-based experimental research projects
- Prepared technical reports for experimental findings

*Structural Engineer, **Purdue Design Build Fly**,* West Lafayette IN

August 2023 - May 2024

- Designed and manufactured structural components for an RC aircraft competing in the AIAA Design/Build/Fly competition
- Collaborated with a multidisciplinary team to design the aircraft fuselage in SolidWorks
- Fabricated wooden aircraft structures and validated manufacturability through iterative design

CAMPUS AND COMMUNITY SERVICE

*Mentee, **Women in Engineering at Purdue**,* West Lafayette IN

August 2023 - Current

- Participates in networking events and seminars that support the success of women and other underrepresented students in engineering
- Develops professional communication and networking skills through mentorship and career development programs

*Volunteer, **Adventure Science Center**,* Nashville TN

May 2024 - August 2024

- Taught and supervised children through hands-on STEM activities, including microscopy and woodworking
- Collaborated with fellow volunteers to create engaging educational experiences

*Resident, **Global Engineering LC**,* West Lafayette IN

August 2023 - December 2023

- Collaborated with students from diverse backgrounds to develop a global perspective and strengthen intercultural communication skills
- Engaged in discussions and activities that strengthened cross-cultural communication and examined engineering challenges from an international perspective

SKILLS, CERTIFICATIONS, AND LANGUAGES

- **Technical Skills:**
 - **Programming:** Python, MATLAB, C, JavaScript, HTML, Git/GitHub, Linux/Command Line
 - **CAD:** SolidWorks, Siemens NX
 - **Computational:** PeleC, AMReX, Ansys Fluent, Ansys Mechanical, VisIt, High-Performance Computing (HPC), SLURM Job Scheduling, MPI Parallel Computing, Adaptive Mesh Refinement (AMR)
 - **Engineering:** Piping and Instrumentation Diagrams (Visio), Product Data Management (PDM) Systems
 - **Productivity:** Microsoft Office, Google Suite, Excel
- **Certifications:** SolidWorks CAD Design Associate (2024), and CornellIX Introduction to Engineering Simulations (FEA & CFD using Ansys) (2025)
- **Language:** German (Intermediate Proficiency)